

MediPulse Executive Summary

Healthcare Operations Intelligence | Python, SQL, Power BI

Metric	Value
Total Patients	55,500
Unique Hospitals	39,876
Unique Doctors	40,341
Total Billing	\$1.42B
Average Billing	\$25,540.29
Average Length of Stay	15.51 days
Emergency Admission Rate	32.92%
Abnormal Test Rate	33.56%
Negative Billing Records Flagged	108

Objective

MediPulse analyzes healthcare encounter data to understand patient demand, billing performance, length of stay, emergency admissions, abnormal test results, and data quality issues.

Methodology

- Cleaned and standardized 55,500 raw healthcare records.
- Converted admission and discharge dates for time-based analysis.
- Engineered length of stay, age group, admission period, stay category, billing per day, emergency flag, abnormal result flag, and repeat visit proxy.
- Flagged negative billing records for audit and prepared dashboard-ready reporting tables.
- Created SQL queries, static visuals, Power BI measures, and dashboard build guidance.

Business Value

- Executive leaders can monitor patient volume, billing, and quality signals from one dashboard.
- Operations teams can identify high-volume conditions and length-of-stay patterns.
- Finance teams can compare insurer billing and review negative billing records.
- Clinical teams can explore abnormal test result patterns by condition and admission type.

Deliverables

- Cleaned dataset: data/cleaned/healthcare_cleaned.csv
- KPI summary: data/cleaned/kpi_summary.csv
- Monthly trend: data/cleaned/monthly_trend.csv
- Condition summary: data/cleaned/condition_summary.csv
- Patient risk scores: data/cleaned/patient_risk_scores.csv
- SQL scripts: sql/01_create_tables.sql and sql/02_business_questions.sql
- Power BI guide: output/pdf/MediPulse_PowerBI_Step_by_Step_Guide.pdf
- Dashboard specification: powerbi/dashboard_spec.md
- Static chart images: images/

Recommended Dashboard Pages

- Executive Overview: KPI cards, admissions trend, conditions, insurer billing, admission type mix.
- Operations: length of stay, top hospitals, weekday admissions, stay categories.
- Financial Performance: insurer billing, condition billing, billing per day, quality review table.
- Clinical Signals: abnormal test rates, medication distribution, age group patterns.

Portfolio Positioning

This project demonstrates a complete analyst workflow: data cleaning, feature engineering, KPI design, SQL analytics, visual storytelling, Power BI planning, quality review, and executive communication.